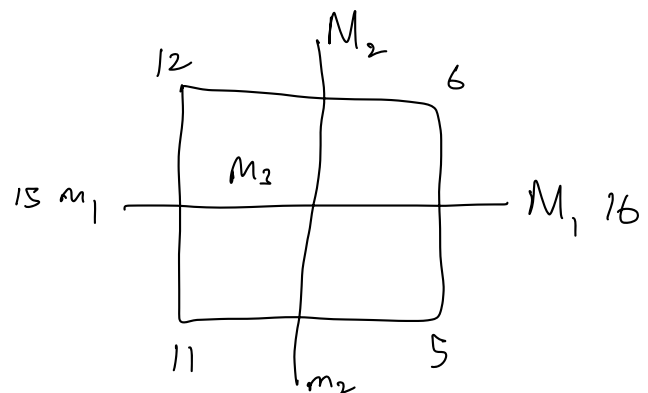
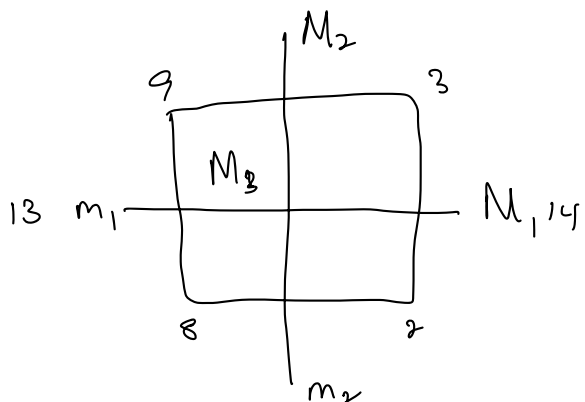
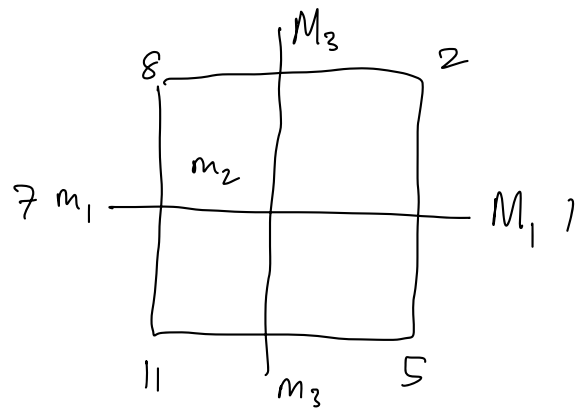
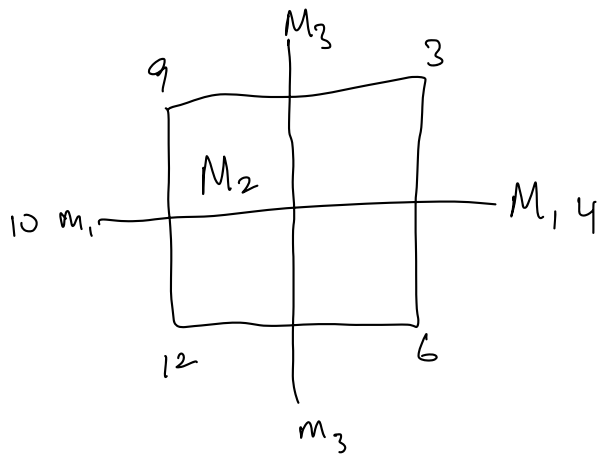
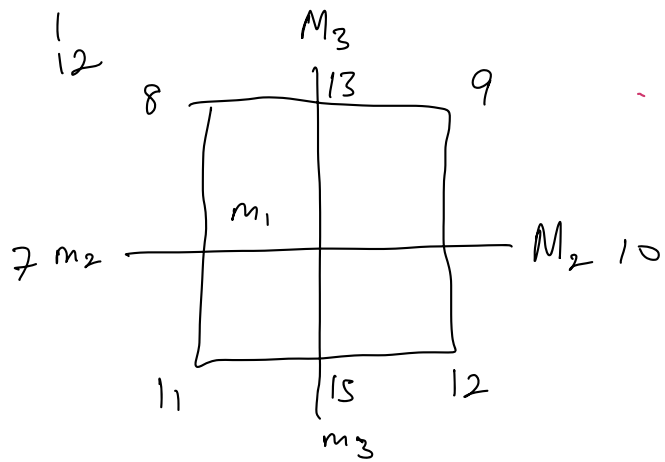
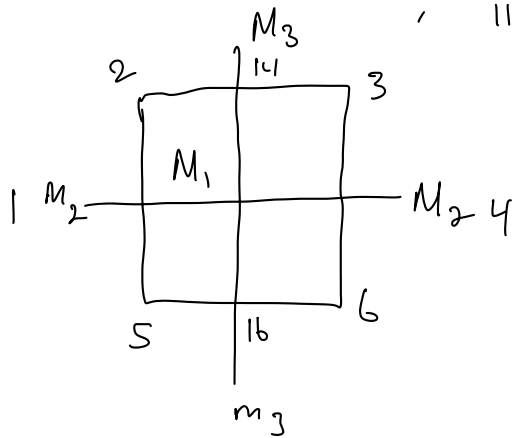
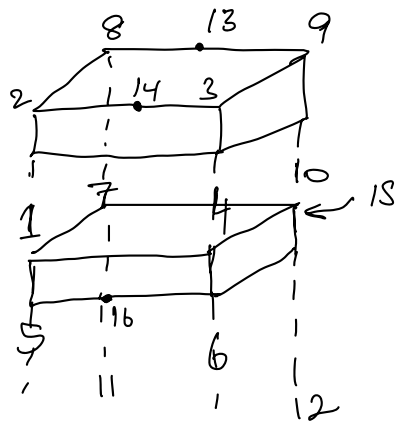
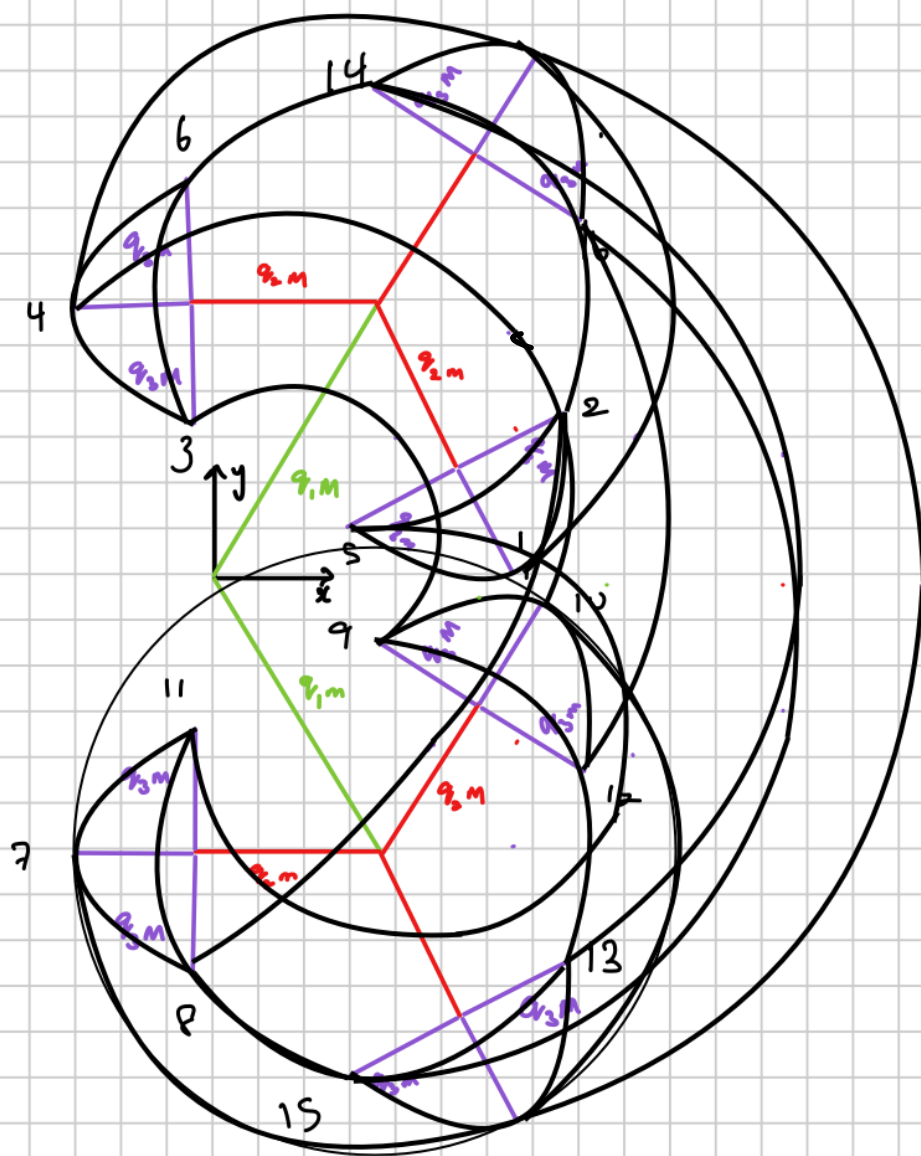


Q1 Imagine the 3 joint space as a cube.





Q2

$$l_1 = l_2 = l_3 = 1m$$

$$\phi = q_1 + q_2 + q_3$$

$$x_e = \begin{bmatrix} x \\ y \\ \phi \end{bmatrix}$$

let w be the position at joint b

$$w = (x_e, y_e) - l_3 [\cos \phi, \sin \phi]$$

$$\cos q_2 = \frac{w_x^2 + w_y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

$$q_1 = \text{atan2}(w_y, w_x) - \text{atan2}(l_2 \sin q_2, l_1 + l_2 \cos q_2)$$

$$q_3 = \phi - q_1 - q_2$$

Using the formula above

$$\text{Posl } x_{e_1} = [2.84, 1.47, 30^\circ]^T$$

$$\omega = (2.84, 1.47) - (\cos 30, \sin 30) = (1.674, 0.97)$$

$$\cos q_2 = \frac{1.674^2 + 0.97^2 - 2}{2}$$

$$q_2 = \cos^{-1}\left(\frac{1.674^2 + 0.97^2 - 2}{2}\right) = \pm 29.36$$

$$\text{when } q_2 = +29.36$$

$$\begin{aligned} q_1 &= \arctan 2(0.97, 1.67) - \arctan 2(\sin q_2, 1 + \cos q_2) \\ &= 15.41^\circ \end{aligned}$$

$$q_3 = 30 - 15.41 - 29.36 = -14.77$$

$$\text{when } q_2 = -29.36$$

$$q_1 = 44.77^\circ$$

$$q_3 = 30 - 44.77 - (-29.36) = 14.59$$

Elbow-down

$$q_1 = 15.41^\circ$$

$$q_2 = 29.36^\circ$$

$$q_3 = -14.77^\circ$$

in white space
but not feasible

Elbow-up

$$q_1 = 44.77^\circ$$

$$q_2 = -29.36^\circ$$

$$q_3 = 14.59^\circ$$

in white space
& feasible

Since $29^\circ < q_1 < 61^\circ$

44.77° is within range

but 15.41° is not

$$\text{Pose } x_{e2} = [1.47 \quad 2.54 \quad 60^\circ]^T$$

$$\omega = (1.47, 2.54) - (\cos 60, \sin 60) = (0.97, 1.674)$$

$$\cos q_2 = \frac{1.674^2 + 0.97^2 - 2}{2}$$

$$q_2 = \cos^{-1}\left(\frac{1.674^2 + 0.97^2 - 2}{2}\right) = \pm 29.36$$

$$\text{when } q_2 = +29.36$$

$$q_1 = \text{atan2}(1.67, 0.97) - \text{atan2}(\sin q_2, 1 + \cos q_2) = 45.23^\circ$$

$$q_3 = 60^\circ - 45.23^\circ - 29.36^\circ = -14.59^\circ$$

$$\text{when } q_2 = -29.36$$

$$q_1 = 74.59$$

$$q_3 = 60 - 74.59 - (-29.36) = 14.77^\circ$$

Elbow - down

$$q_1 = 45.23^\circ$$

$$q_2 = 29.36^\circ$$

$$q_3 = -14.59^\circ$$

In white space
feasible

Elbow - up

$$q_1 = 74.59^\circ$$

$$q_2 = -29.36^\circ$$

$$q_3 = 14.77^\circ$$

In white space
not feasible

Since $29 < q_1 < 61$

45.23 is within range

but 74.59 is not

Pose x_{e_1} Instructions

Elbow-up

$$\begin{matrix} [44.77, -29.36, 14.59] \\ \theta_1 \quad \theta_2 \quad \theta_3 \end{matrix}$$

Pose x_{e_2} Instructions

Elbow-down

$$\begin{matrix} [45.23, 29.36, -14.59] \\ \theta_1 \quad \theta_2 \quad \theta_3 \end{matrix}$$

